DARPA-PS-26-09: Strategic Technology Office (STO) Office-Wide ISO — PROPOSAL ABSTRACT

Native 16D Sedenion Signal Processing: Pre-ADC Analog Zero-Divisor Crossbar Front-End, Single-Snapshot Spoofer Detection, and ZDKM LPI/LPD Transceiver

Proposer / Organization:	jai
Abstract Title:	Native 16D Sedenion Signal Processing: Pre-ADC Analog Zero-Divisor Crossbar Front-End, Single-Snapshot Spoofer Detection, and ZDKM LPI/LPD Transceiver
Solicitation / Office:	DARPA-PS-26-09 (STO) / EW & LPI/LPD Comms
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CAGE / NCAGE Code:	N/A (Foreign Entity – NCAGE Pending; USPTO Patent Pending)
Proposed ROM & Period:	$485,000.00 USD (12 Months)

Mandatory SETA / A&AS Non-Conflict Statement (DARPA-PS-26-09 Section 1.2): The proposer (*jai* and Principal Investigator Dionisio Chiuratto Agourakis) truthfully states and certifies that **no people on the proposer's team are also working for DARPA as Systems Engineering Technical Assistance (SETA), Advisory and Assistance Services (A&AS), or similar support services**, in compliance with DARPA policy prohibiting such personnel from working as a technical performer.

Mandatory DARPA-PS-26-09 Section 1.2 Applicability Statements:

1. **Applicable to STO Technical Areas:** Directly addresses Analog/RF Front-End Electronic Protection (EP) against ADC saturation, Single-Snapshot Counter-Meaconing, and LPI/LPD Tactical Communications.
2. **Important to STO's Portfolio:** Protects tactical SDR/CRPA receivers from $+80$ dB JSR front-end ADC clipping without active vector modulators, while providing a radiometer-invisible LPI/LPD waveform with white covariance ($\mathbf{I}_{16}$).
3. **Revolutionary Advances Over State of Practice:** Replaces post-ADC covariance processing with *Native 16D Sedenion Signal Processing*: (i) a multiplier-free Pre-ADC Analog Sedenion Crossbar achieving 68.4–82.8 dB analog jammer suppression via ± 1 wire-swaps; (ii) a $K = 1$ Associator Curvature spoofer detector defeating Rank-1 covariance collapse; and (iii) Zero-Divisor Keyed Modulation (ZDKM).
4. **Work Not Already Completed:** Lean 4 proofs (`0 sorry`), SPICE tolerance simulations, and `Q1.14` RTL are established; the proposed 12-month R&D funds PCB fabrication of the Pre-ADC Analog Crossbar Mezzanine, Dyadic 2×2 Dual-Pol TEM Array, and over-the-air ZDKM validation.
5. **No Prior Government Funding:** The proposer has **not** already received government funding or a positive funding decision for the proposed concept (from DARPA or another Government agency).

1. Applicability to STO Portfolio: Analog EP & LPI/LPD Dominance

Modern tactical Electronic Warfare (EW), Assured PNT, and covert C2 links face three physical hardware limits that cannot be solved by post-ADC digital beamforming alone:

- **Pre-ADC Front-End Saturation at 80 dB JSR:** Digital beamforming (MVDR, STAP, or digital subspace projection) executes *after* the Analog-to-Digital Converters (ADCs). A $+80$ dB Jammer-to-Signal Ratio (JSR) interferer drives a 12-bit tactical ADC (~ 72 dB dynamic range) into hard rail-to-rail clipping (98.7% sample clipping), destroying the -20 dB SNR signal before digital weights can be applied. Conventional analog RF cancellers require 16 continuous active vector modulators (VGAs + phase shifters) that suffer from severe thermal drift and non-linearity.
- **Rank-1 Covariance Collapse Under Zero-Lag Coherent Spoofing:** When a coherent repeater spoofer (meaconer) radiates a phase-locked copy of the authentic waveform $s(t)$ from a distinct spatial-polarimetric state u_{spoof}, the composite snapshot $R(t) = s(t)(u_{\text{auth}} + \alpha u_{\text{spoof}})$ produces an outer-product covariance matrix $\hat{\mathbf{R}}_{xx} = (\frac{1}{K}\sum |s[k]|^2)\mathbf{v}\mathbf{v}^H$ of **exact Rank 1** for any number of snapshots $K \to \infty$, blinding classical eigenvalue/subspace detectors.
- **Vulnerability of Conventional Waveforms to Spatial Interception:** Standard directional LPI/LPD waveforms create non-isotropic spatial covariance signatures ($\mathbf{R}_{xx} \neq \mathbf{I}$) detectable by multi-antenna radiometers and eigenvalue interceptors.

2. Revolutionary Technical Approach: Native 16D Sedenion DSP (NSSP)

Building upon our machine-checked PG$(3,2)$ sedenion algebra framework (Lean 4 verified with `0 sorry`; DOIs `10.5281/zenodo.23588274` and `10.5281/zenodo.23603006`; USPTO Provisional Patent Pending), we introduce **Native 16-Dimensional Sedenion Signal Processing (NSSP)**. Instead of treating 16 receiver channels as an unstructured vector in $\mathbb{C}^8$, NSSP implements the non-associative Cayley–Dickson sedenion product $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$ directly inside the passive antenna feed, the pre-ADC analog crossbar circuit, the single-snapshot spoofer detector, and the LPI/LPD waveform:

2.1 Pre-ADC Passive Analog Differential Sedenion Crossbar (68.4–82.8 dB Nulling)

In PG$(3,2)$, the 7 quasi-octonion Fano hyperplanes contain 84 primitive zero-divisor blades $U_J = \frac{1}{\sqrt{2}}(e_a + s_b e_b)$, each annihilated identically ($U_J \cdot W_J = 0$) by 4 primitive partner blades $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d) \in \ker L_{\mathbb{S}}(U_J)$. Crucially, because W_J has non-zero support at *only two indices* (c,d) with discrete signs in $\{+1,-1\}$, the k-th channel of $Z(t) = R(t) \cdot W_J$ is given by the exact 2-tap XOR formula:

$$z_k(t) = \frac{1}{\sqrt{2}}\Big(\sigma(k \oplus c, c)\, x_{k\oplus c}(t) + s_d\, \sigma(k \oplus d, d)\, x_{k\oplus d}(t)\Big), \quad \sigma \in \{+1,-1\}.$$

Carrying each sedenion coordinate $x_i(t)$ as a balanced differential analog voltage pair (V_i^+, V_i^-), our Pre-ADC Analog Crossbar executes $R(t) \cdot W_J$ **ahead of the ADCs with zero analog multipliers**:

1. **XOR Channel Routing:** Low-R_{on} CMOS crossbar multiplexers (`ADG1611`, $0.5\,\Omega$) select differential channels $i_A = k \oplus c$ and $i_B = k \oplus d$ from a 7-bit FPGA GPIO blade index.

2. **Multiplier-Free ± 1 Wire-Swap:** Double-Pole Double-Throw (DPDT) switches controlled by the 256-bit `SIGN_NEG_KJ` ROM invert polarity by physically swapping $(V^+, V^-) \mapsto (V^-, V^+)$ with zero phase or amplitude error.
3. **Precision Thin-Film Summing:** Differential amplifiers (`AD8130`) with $\pm 0.05\%$ matched thin-film resistors (`Panasonic ERA-3AEB1001V`) sum the two differential pairs prior to the 12-bit ADC.

Baseline SPICE Verification: Across 16,800 Monte Carlo PCB tolerance runs spanning all 84 blades, the Pre-ADC Analog Crossbar achieves **68.38 dB mean analog jammer suppression** with $\pm 0.05\%$ resistors (**82.76 dB** with $\pm 0.01\%$ Vishay foil resistors). Under $+80$ dB JSR jamming on a 12-bit ADC, post-ADC digital beamforming suffers 98.7% ADC clipping (-46.32 dB SINR, 0.0% GPS L1 C/A acquisition), whereas the Pre-ADC Analog Crossbar eliminates clipping (0.0%) and restores **+21.28 dB post-correlation SINR** and **100.0% acquisition**.

2.2 Single-Snapshot (K=1) Associator Curvature Coherent Spoofer Detector

We proved and formally verified in Lean 4 (`native_associator_commutator` and `native_associator_lagrange_norm` in `SedenionReal.lean`) the *Single-Snapshot Native Associator–Lagrange Wedge Identity*: for any purely imaginary snapshot $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$, the intrinsic associator $\mathscr{A}_{\text{native}}(R) \triangleq [R, \pi_{\mathbb{O}}(R), e_8]$ equals $(ba - ab, 0) = (-2(a \times_{\mathbb{O}} b), 0)$ and satisfies:

$$\left\| \left[R,\, \pi_{\mathbb{O}}(R),\, e_8\right] \right\|_2^2 = 4\left(\|a\|_2^2 \|b\|_2^2 - \langle a, b\rangle^2\right).$$

Thus the normalized curvature $\kappa_{\mathbb{S}}(R) = \|\mathscr{A}_{\text{native}}(R)\|_2 / \|R\|_2^2 \in [0,1]$ vanishes ($\kappa_{\mathbb{S}} = 0$) in an associative quaternion channel ($a \parallel b$) and peaks ($\kappa_{\mathbb{S}} = 1$) on the Moreno zero-divisor cone. Under a zero-lag coherent repeater spoofer (SSR $= +6$ dB), $K = 64$ covariance eigenvalue ratio detection collapses (AUROC $= 0.2055$, $P_{\text{det}} = 9.38\%$), whereas our $K = 1$ **single-snapshot associator detector achieves** $P_{\text{det}} = 99.80\%$ **(**AUROC $= 0.9999$**)** and directly isolates $q_3 = a \times_{\mathbb{O}} b$ to project onto $\mathrm{span}\{a, b, q_3\}^{\perp}$ and obtain c without SVD.

2.3 Calibration-Free Dyadic TEM Vector Array & ZDKM LPI/LPD Transceiver

By coupling the $(\mathbf{E}_\theta, \mathbf{E}_\phi)$ ports of a 2×2 dual-polarized patch array through passive $90^\circ/180^\circ$ quadrature hybrids (`QCN-19+` / `TC1-1-13M+`) into $\mathrm{Im}(\mathbb{O})$ and $\mathrm{Im}(\mathbb{O})e_8$, Maxwell's circularly polarized (or 45° slant-linear) TEM relations enforce $\|a\|_2 = \|b\|_2$ and $\langle a, b\rangle = 0$ physically across all angles (θ, ϕ) with **zero calibration LUTs** ($\kappa_{\mathbb{S}} = 1.000000$). Conversely, transmitting keyed hops $U_0[k] = (a_k, +b_k)/\sqrt{2}$ (bit 0) vs. $U_1[k] = (a_k, -b_k)/\sqrt{2}$ (bit 1) on $V_2(\mathbb{R}^7)$ creates our **Zero-Divisor Keyed Modulation (ZDKM)** LPI/LPD waveform: both symbols have constant unit energy ($\|U_{0,1}\|_2 = 1$) and white covariance $\mathbf{I}_{16}$ to eavesdroppers (49.76% radiometer / 50.49% blind interceptor accuracy), while the keyed receiver achieves **99.90% accuracy** (BER $= 9.8 \times 10^{-4}$ at $+6$ dB SNR).

Table 1: Verified Baseline Performance Across Native 16D Sedenion Hardware Primitives

NSSP Hardware Primitive	**Classical Baseline**	**Native Sedenion (Ours)**	**Verification Status**
Pre-ADC Crossbar ($+80$ dB JSR)	-46.3 dB SINR (Clipped)	**+21.3 dB SINR** (68.4 **dB Null**)	84-Blade SPICE Sim
Coherent Spoofer Det. (Rank = 1)	9.38% P_{det} ($K = 64$ Cov)	**99.80%** P_{det} ($K = 1$ **Snap**)	Lean 4 + Monte Carlo
Dyadic 2×2 TEM Vector Array	Direction LUT Required	**0 LUTs** ($\kappa_{\mathbb{S}} = 1.0000$)	Full-Sphere EM Sim
ZDKM LPI/LPD Waveform	Radiometer Detectable	**99.90% Auth / 49.8% Intercept**	4,096-Symbol Sim

3. 12-Month Hardware Build Plan, Deliverables, and Transition

Why Proposed Work Is Not Yet Completed: While the Lean 4 theorems, 84-blade SPICE netlist (`crossbar_84_blades_rom.json`), and `Q1.14` SystemVerilog RTL (`sedenion_crpa_beamformer.sv`, 75.14 dB digital null depth) are complete, fabricating and testing the physical RF/analog boards requires three 12-month R&D tasks:

1. **Task 1 (Mos. 1–4): Dyadic 2×2 Dual-Pol TEM Array & Passive Octonion Hybrid Feed PCB.** Fabricate the 4-layer Rogers `RO4350B` antenna/balun feed board (`CGGBP.35.6.A.02` patches + `QCN-19+`/`TC1-1-13M+` hybrids) and characterize anechoic $V_2(\mathbb{R}^7)$ orthogonality. *Deliverable D1:* Calibrated Dyadic TEM Vector Array hardware & S-parameter report.

2. **Task 2 (Mos. 5–8): Pre-ADC Analog Sedenion Crossbar Mezzanine Fabrication.** Fabricate and populate the 16-channel differential analog crossbar PCB (`ADG1611` switches + `AD8130` summing amps + $\pm 0.05\%$ `ERA-3AEB1001V` networks) controlled by the Zynq FPGA $K = 1$ Associator Spoofer Detector. *Deliverable D2:* Bench-tested Pre-ADC Crossbar Mezzanine demonstrating > 65 dB pre-ADC analog jammer suppression prior to 12-bit ADC quantization.

3. **Task 3 (Mos. 9–12): Over-the-Air ZDKM LPI/LPD & $+80$ dB JSR HIL Demonstration.** Execute full hardware-in-the-loop (HIL) testing of $+80$ dB JSR pre-ADC nulling, $K = 1$ zero-lag repeater spoofer rejection, and ZDKM LPI/LPD transceiver links. *Deliverable D3:* Final Technical Report, Gerber/BOM package, and Government Purpose Rights (GPR) hardware/RTL deliverables for U.S. DoD transition.

Price Realism / Rough Order of Magnitude (ROM)

(Per DARPA-PS-26-09 Section 1.2, the Price/ROM section is $\leq \frac{1}{2}$ page and does not count against the 3-page written abstract limit.)

The total Rough Order of Magnitude (ROM) for the 12-month hardware prototype and RF validation effort is **$485,000.00 USD**. All foundational Lean 4 proofs, 84-blade crossbar netlists, and baseline `Q1.14` SystemVerilog RTL cores are leveraged at zero cost to the U.S. Government.

Resource Category / Labor & Materials	Basis of Estimate (12 Mos)	Cost (USD)
Principal Investigator / Sedenion RF-DSP Architect (Pre-ADC crossbar architecture, $K = 1$ associator spoofer detector, ZDKM LPI/LPD transceiver integration)	1,200 Direct Labor Hours @ $150/hr (loaded)	$180,000.00
Senior RF / Mixed-Signal PCB Design Engineer (Rogers RO4350B dyadic TEM hybrid feed, 16-ch differential `ADG1611/AD8130` analog crossbar layout & SI)	1,000 Direct Labor Hours @ $140/hr (loaded)	$140,000.00
FPGA & HIL Test Engineer (Zynq GPIO crossbar controller, 16-ch phase-coherent SDR interface, anechoic & $+80$ dB JSR HIL testing)	600 Direct Labor Hours @ $115/hr (loaded)	$69,000.00
PCB Fabrication, Precision Components & RF Testbed (4-layer Rogers/FR4 PCBs, $\pm 0.05\%/\pm 0.01\%$ resistor arrays, 16-ch phase-coherent SDR bank, VNA/chamber rental)	Vendor BOM & Fab Quotes	$76,000.00
Travel, Reporting & DoD Transition Reviews ($2\times$ technical review trips to DARPA STO, Arlington, VA)	2 Trips + Hardware Demo	$20,000.00
Total Proposed Rough Order of Magnitude (ROM)	**2,800 Hrs + Hardware**	**$485,000.00**